

analysis_multilevel.out

Mplus VERSION 8.11
MUTHEN & MUTHEN
04/16/2026 8:03 PM

INPUT INSTRUCTIONS

TITLE: Two-Level Multilevel Bayesian Analysis (Students in Schools);

DATA:

FILE IS mplus_multilevel_data.dat;

VARIABLE:

NAMES ARE REGN SCHL SPED;
USEVARIABLES ARE SPED SCHL
AFAM_F AFAM_M AMIND_F AMIND_M ASIAN_F ASIAN_M
HAW_F HAW_M HISP_F HISP_M MULTI_F MULTI_M WHT_M;

CATEGORICAL IS SPED;
CLUSTER IS SCHL; ! Defines the nesting (Level 2)
MISSING ARE ALL (-99);
WITHIN = AFAM_F AFAM_M AMIND_F AMIND_M ASIAN_F ASIAN_M
HAW_F HAW_M HISP_F HISP_M MULTI_F MULTI_M WHT_M;

DEFINE:

! Dummies (White_Female Category 13 is reference)
AFAM_F = 0; IF (REGN EQ 1) THEN AFAM_F = 1;
AFAM_M = 0; IF (REGN EQ 2) THEN AFAM_M = 1;
AMIND_F = 0; IF (REGN EQ 3) THEN AMIND_F = 1;
AMIND_M = 0; IF (REGN EQ 4) THEN AMIND_M = 1;
ASIAN_F = 0; IF (REGN EQ 5) THEN ASIAN_F = 1;
ASIAN_M = 0; IF (REGN EQ 6) THEN ASIAN_M = 1;
HAW_F = 0; IF (REGN EQ 7) THEN HAW_F = 1;
HAW_M = 0; IF (REGN EQ 8) THEN HAW_M = 1;
HISP_F = 0; IF (REGN EQ 9) THEN HISP_F = 1;
HISP_M = 0; IF (REGN EQ 10) THEN HISP_M = 1;
MULTI_F = 0; IF (REGN EQ 11) THEN MULTI_F = 1;
MULTI_M = 0; IF (REGN EQ 12) THEN MULTI_M = 1;
WHT_M = 0; IF (REGN EQ 14) THEN WHT_M = 1;

ANALYSIS:

TYPE = TWOLEVEL; ! Enables Multilevel modeling
ESTIMATOR = BAYES;
PROCESSORS = 4;
BITERATIONS = (5000);
BCONVERGENCE = 0.05;

MODEL:

%WITHIN% ! Level 1 (Student Level)
SPED ON AFAM_F AFAM_M AMIND_F AMIND_M ASIAN_F ASIAN_M
HAW_F HAW_M HISP_F HISP_M MULTI_F MULTI_M WHT_M;

%BETWEEN% ! Level 2 (School Level)
SPED; ! Estimates the random intercept (school-level variance)

OUTPUT:

TECH1 TECH8 STDYX;

*** WARNING

One or more individual-level variables have no variation within a cluster for the following clusters.

Variable Cluster IDs with no within-cluster variation

```
SPED 765 609 888 739 737 527 108 530 679 689 539 768 615 763 645 512 529 750 650
      630 538
AFAM_F 609 888 566
AFAM_M 609 888 572
AMIND_F 765 609 888 739 737 527 574 4060 530 679 354 539 768 615 663 645 566 637 514
      655 745 522 512 540 356 535 526 13 529 628 750 543 545 728 584 456 330 240 779
      776 568
AMIND_M 765 609 888 739 737 527 108 574 4060 689 354 539 768 572 37 565 566 537 514
      522 4065 356 535 526 623 728 244 314 326 260 322 571 21 710 240 242 252 776
      575
ASIAN_F 765 609 888 739 737 527 108 574 4060 530 679 689 354 768 615 565 763 645 566
      537 655 356 358 575
ASIAN_M 765 609 888 739 737 527 574 4060 689 768 572 615 565 566 537 655 358 4045 581
      575
HAW_F 765 609 888 739 737 527 108 574 4060 530 679 689 354 539 768 615 37 763 663
      645 566 637 537 514 655 745 522 4065 512 540 567 570 535 13 529 516 613 4090
      623 750 358 650 510 362 630 22 638 4075 728 244 640 658 324 538 618 260 350
      132 662 759 339 130 274 660 668 721 240 722 4045 732 634 712 669 250 775 720
      280 581 672 270 724 690 180 743 692 294 190 664 284 588
HAW_M 765 609 888 739 737 527 108 574 530 689 354 539 768 572 615 37 565 763 645 637
      537 36 514 655 522 570 356 535 13 529 516 613 4090 450 691 628 750 543 650 510
      362 29 33 22 728 310 244 658 740 324 156 326 264 260 670 350 573 246 418 662
      14 339 749 456 660 240 128 242 722 4045 732 10 254 686 620 669 280 730 282 312
      586 575 733 621 743 692 766 734 294 290 665 664 284 748 588
HISP_F 765 609 888 739 737 108 566
HISP_M 765 572
MULTI_F 765 609 888 739 737 108 354 539 763 566 514 4075
MULTI_M 609 888 527 108 354 539 572 566 567
WHT_M 572 565
```

1 WARNING(S) FOUND IN THE INPUT INSTRUCTIONS

Two-Level Multilevel Bayesian Analysis (Students in Schools);

SUMMARY OF ANALYSIS

Number of groups 1

Number of observations 1389102

Number of dependent variables 1

Number of independent variables 13

Number of continuous latent variables 0

Observed dependent variables

Binary and ordered categorical (ordinal)

SPED

Observed independent variables

AFAM_F AFAM_M AMIND_F AMIND_M ASIAN_F ASIAN_M

HAW_F HAW_M HISP_F HISP_M MULTI_F MULTI_M

WHT_M

Variables with special functions

Cluster variable SCHL

Within variables

AFAM_F AFAM_M AMIND_F AMIND_M ASIAN_F ASIAN_M

HAW_F HAW_M HISP_F HISP_M MULTI_F MULTI_M

WHT_M

Estimator BAYES
Specifications for Bayesian Estimation
Point estimate MEDIAN
Number of Markov chain Monte Carlo (MCMC) chains 2
Random seed for the first chain 0
Starting value information UNPERTURBED
Algorithm used for Markov chain Monte Carlo GIBBS(PX1)
Convergence criterion 0.500D-01
Maximum number of iterations 50000
K-th iteration used for thinning 1
Link PROBIT

Input data file(s)
 mplus_multilevel_data.dat
Input data format FREE

SUMMARY OF DATA

Number of clusters 253

Size (s) Cluster ID with Size s

6	765
7	609
15	888
34	739
60	737
108	527
113	108
208	574
215	4060 530
316	679
321	689 354
339	539
369	768
377	572
378	615
387	37
389	565
398	763
432	663
553	645
613	566
615	633
638	637
739	537
818	36
826	514
895	655
897	745
918	522
1104	4065
1118	512
1124	540
1164	567
1197	570
1258	356
1317	535
1356	526
1370	13
1372	529
1424	516
1442	613
1443	4090
1502	450

1559 691
1572 623
1699 628
1711 753
1749 750
1863 358
1896 543
1943 650
1962 510
2147 27
2306 362
2328 29
2371 630
2397 33
2668 22
2851 638
2887 4075
2931 545
2949 34
2974 728
3001 642
3011 310
3030 244
3056 640
3114 35
3128 658
3138 636
3159 584
3227 334
3282 740
3360 324
3419 538
3502 332
3511 635
3516 632
3529 156
3556 4055
3624 420
3641 314
3654 4040
3674 124
3677 248
3727 618
3766 326
3788 410
3798 4080
3904 264
3935 612
3944 752
4032 260
4053 670 350
4092 266
4100 573
4131 132
4160 246
4185 760
4212 418
4278 662
4291 759
4321 322
4326 14
4372 339
4401 256
4411 571
4475 761
4477 318
4496 762

4529 718
4532 749
4544 352
4647 21
4748 456
4760 616
4855 130
4871 112
4884 274
4898 710 660
4914 680
4964 696
5010 330
5034 432
5036 682
5040 668
5102 721
5132 240
5151 611
5165 128
5177 4050
5190 684
5271 622
5298 242
5302 110
5387 320
5392 722
5396 4045
5402 252
5464 732
5482 673
5530 10
5541 254
5584 681
5595 686
5612 321
5645 271
5731 412
5783 578
5802 634
5810 620
5888 28
6007 20
6018 779
6055 776
6059 712
6095 669
6108 250
6140 756
6173 775
6206 720
6289 736
6294 327
6310 783
6339 11
6416 280
6423 581
6460 672
6491 270
6503 782
6508 724
6556 730
6743 780
6751 282
6942 610
6972 312
7021 32

7041 586
7066 575
7176 733
7230 474
7294 476
7312 176
7335 374
7352 690
7363 741
7370 742
7681 579
7793 276
7847 16
7878 180
7911 621
7973 743
7982 692
8010 766
8183 427
8223 392
8234 734
8246 764
8297 738
8321 625
8419 151
8480 178
8558 376
8570 17
8593 294
8596 644
8610 190
8655 292
8718 576
8742 290
8791 170
8796 26
8818 646
8938 665
9213 664
9347 286
9399 372
9419 568
9508 23
9670 746
9704 295
9732 685
9869 261
10132 390
10246 675
10351 470
10808 196
10836 194
10952 66
11186 678
11188 44
11205 422
11560 394
11773 284
11971 652
12556 587
13222 770
15221 18
15680 726
15796 80
15882 24
15932 688
17144 585

17230 748
18018 648
20859 626
21715 490
24490 588

COVARIANCE COVERAGE OF DATA

Minimum covariance coverage value 0.100

Number of missing data patterns 1

PROPORTION OF DATA PRESENT

Covariance Coverage

SPED AFAM_F AFAM_M AMIND_F AMIND_M

SPED	1.000				
AFAM_F	1.000	1.000			
AFAM_M	1.000	1.000	1.000		
AMIND_F	1.000	1.000	1.000	1.000	
AMIND_M	1.000	1.000	1.000	1.000	1.000
ASIAN_F	1.000	1.000	1.000	1.000	1.000
ASIAN_M	1.000	1.000	1.000	1.000	1.000
HAW_F	1.000	1.000	1.000	1.000	1.000
HAW_M	1.000	1.000	1.000	1.000	1.000
HISP_F	1.000	1.000	1.000	1.000	1.000
HISP_M	1.000	1.000	1.000	1.000	1.000
MULTI_F	1.000	1.000	1.000	1.000	1.000
MULTI_M	1.000	1.000	1.000	1.000	1.000
WHT_M	1.000	1.000	1.000	1.000	1.000

Covariance Coverage

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

ASIAN_F	1.000				
ASIAN_M	1.000	1.000			
HAW_F	1.000	1.000	1.000		
HAW_M	1.000	1.000	1.000	1.000	
HISP_F	1.000	1.000	1.000	1.000	1.000
HISP_M	1.000	1.000	1.000	1.000	1.000
MULTI_F	1.000	1.000	1.000	1.000	1.000
MULTI_M	1.000	1.000	1.000	1.000	1.000
WHT_M	1.000	1.000	1.000	1.000	1.000

Covariance Coverage

HISP_M MULTI_F MULTI_M WHT_M

HISP_M	1.000			
MULTI_F	1.000	1.000		
MULTI_M	1.000	1.000	1.000	
WHT_M	1.000	1.000	1.000	1.000

UNIVARIATE PROPORTIONS AND COUNTS FOR CATEGORICAL VARIABLES

SPED

Category 1 0.838 1164590.000
Category 2 0.162 224512.000

UNIVARIATE SAMPLE STATISTICS

UNIVARIATE HIGHER-ORDER MOMENT DESCRIPTIVE STATISTICS

Variable/	Mean/	Skewness/	Minimum/	% with Percentiles
Sample Size	Variance	Kurtosis	Maximum	Min/Max 20%/60% 40%/80% Median
AFAM_F	0.149	1.967	0.000	85.06% 0.000 0.000 0.000
1389102.000	0.127	1.870	1.000	14.94% 0.000 0.000
AFAM_M	0.155	1.904	0.000	84.48% 0.000 0.000 0.000
1389102.000	0.131	1.626	1.000	15.52% 0.000 0.000
AMIND_F	0.002	22.516	0.000	99.80% 0.000 0.000 0.000
1389102.000	0.002	504.951	1.000	0.20% 0.000 0.000
AMIND_M	0.002	21.772	0.000	99.79% 0.000 0.000 0.000
1389102.000	0.002	472.029	1.000	0.21% 0.000 0.000
ASIAN_F	0.020	6.942	0.000	98.05% 0.000 0.000 0.000
1389102.000	0.019	46.197	1.000	1.95% 0.000 0.000
ASIAN_M	0.021	6.738	0.000	97.93% 0.000 0.000 0.000
1389102.000	0.020	43.403	1.000	2.07% 0.000 0.000
HAW_F	0.001	39.075	0.000	99.93% 0.000 0.000 0.000
1389102.000	0.001	1524.849	1.000	0.07% 0.000 0.000
HAW_M	0.001	38.423	0.000	99.93% 0.000 0.000 0.000
1389102.000	0.001	1474.343	1.000	0.07% 0.000 0.000
HISP_F	0.087	2.933	0.000	91.31% 0.000 0.000 0.000
1389102.000	0.079	6.602	1.000	8.69% 0.000 0.000
HISP_M	0.090	2.870	0.000	91.02% 0.000 0.000 0.000
1389102.000	0.082	6.235	1.000	8.98% 0.000 0.000
MULTI_F	0.021	6.694	0.000	97.91% 0.000 0.000 0.000
1389102.000	0.020	42.816	1.000	2.09% 0.000 0.000
MULTI_M	0.021	6.685	0.000	97.90% 0.000 0.000 0.000
1389102.000	0.021	42.690	1.000	2.10% 0.000 0.000
WHT_M	0.224	1.326	0.000	77.63% 0.000 0.000 0.000
1389102.000	0.174	-0.241	1.000	22.37% 0.000 1.000

THE MODEL ESTIMATION TERMINATED NORMALLY

USE THE FBITERATIONS OPTION TO INCREASE THE NUMBER OF ITERATIONS BY A FACTOR OF AT LEAST TWO TO CHECK CONVERGENCE AND THAT THE PSR VALUE DOES NOT INCREASE.

MODEL FIT INFORMATION

Number of Free Parameters 15

Bayesian Posterior Predictive Checking using Chi-Square

95% Confidence Interval for the Difference Between
the Observed and the Replicated Chi-Square Values

-18.366 15.555

Posterior Predictive P-Value 0.510

MODEL RESULTS

Posterior One-Tailed 95% C.I.
Estimate S.D. P-Value Lower 2.5% Upper 2.5% Significance

Within Level

SPED ON

AFAM_F	0.158	0.005	0.000	0.149	0.168	*
AFAM_M	0.512	0.004	0.000	0.504	0.521	*
AMIND_F	0.039	0.034	0.129	-0.025	0.104	
AMIND_M	0.345	0.029	0.000	0.288	0.400	*
ASIAN_F	-0.474	0.016	0.000	-0.508	-0.443	*
ASIAN_M	-0.119	0.012	0.000	-0.143	-0.095	*
HAW_F	-0.051	0.059	0.197	-0.166	0.062	
HAW_M	0.071	0.057	0.111	-0.043	0.180	
HISP_F	0.005	0.006	0.179	-0.006	0.017	
HISP_M	0.305	0.005	0.000	0.294	0.316	*
MULTI_F	0.025	0.010	0.007	0.005	0.046	*
MULTI_M	0.368	0.009	0.000	0.350	0.386	*
WHT_M	0.333	0.004	0.000	0.325	0.341	*

Between Level

Thresholds

SPED\$1	0.793	0.088	0.000	0.623	0.963	*
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Variances

SPED	1.890	0.187	0.000	1.551	2.289	*
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STANDARDIZED MODEL RESULTS

STDYX Standardization

	Estimate	S.D.	P-Value	Lower 2.5%	Upper 2.5%	Significance
Posterior One-Tailed 95% C.I.						

Within Level

SPED ON

AFAM_F	0.055	0.002	0.000	0.052	0.059	*
AFAM_M	0.182	0.002	0.000	0.179	0.185	*
AMIND_F	0.002	0.001	0.129	-0.001	0.005	
AMIND_M	0.015	0.001	0.000	0.013	0.018	*
ASIAN_F	-0.064	0.002	0.000	-0.069	-0.060	*
ASIAN_M	-0.017	0.002	0.000	-0.020	-0.013	*
HAW_F	-0.001	0.001	0.197	-0.004	0.002	
HAW_M	0.002	0.001	0.111	-0.001	0.005	
HISP_F	0.001	0.002	0.179	-0.002	0.005	
HISP_M	0.085	0.002	0.000	0.082	0.088	*
MULTI_F	0.004	0.001	0.007	0.001	0.006	*
MULTI_M	0.052	0.001	0.000	0.049	0.054	*
WHT_M	0.136	0.002	0.000	0.133	0.139	*

Between Level

Thresholds

SPED\$1	0.578	0.071	0.000	0.442	0.716	*
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Variances

SPED	1.000	0.000	0.000	1.000	1.000	
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R-SQUARE

Within Level

Variable	Estimate	S.D.	P-Value	Lower 2.5%	Upper 2.5%
Posterior One-Tailed 95% C.I.					

SPED	0.042	0.001	0.000	0.041	0.044
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TECHNICAL 1 OUTPUT

PARAMETER SPECIFICATION FOR WITHIN

TAU

SPED\$1

0

NU

SPED AFAM_F AFAM_M AMIND_F AMIND_M

0 0 0 0 0

NU

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

0 0 0 0 0

NU

HISP_M MULTI_F MULTI_M WHT_M

0 0 0 0

LAMBDA

SPED AFAM_F AFAM_M AMIND_F AMIND_M

SPED 0 0 0 0 0

AFAM_F 0 0 0 0 0

AFAM_M 0 0 0 0 0

AMIND_F 0 0 0 0 0

AMIND_M 0 0 0 0 0

ASIAN_F 0 0 0 0 0

ASIAN_M 0 0 0 0 0

HAW_F 0 0 0 0 0

HAW_M 0 0 0 0 0

HISP_F 0 0 0 0 0

HISP_M 0 0 0 0 0

MULTI_F 0 0 0 0 0

MULTI_M 0 0 0 0 0

WHT_M 0 0 0 0 0

LAMBDA

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

SPED 0 0 0 0 0

AFAM_F 0 0 0 0 0

AFAM_M 0 0 0 0 0

AMIND_F 0 0 0 0 0

AMIND_M 0 0 0 0 0

ASIAN_F 0 0 0 0 0

ASIAN_M 0 0 0 0 0

HAW_F 0 0 0 0 0

HAW_M 0 0 0 0 0

HISP_F 0 0 0 0 0

HISP_M 0 0 0 0 0

MULTI_F 0 0 0 0 0

MULTI_M 0 0 0 0 0

WHT_M 0 0 0 0 0

LAMBDA

HISP_M MULTI_F MULTI_M WHT_M

```

SPED 0 0 0 0
AFAM_F 0 0 0 0
AFAM_M 0 0 0 0
AMIND_F 0 0 0 0
AMIND_M 0 0 0 0
ASIAN_F 0 0 0 0
ASIAN_M 0 0 0 0
HAW_F 0 0 0 0
HAW_M 0 0 0 0
HISP_F 0 0 0 0
HISP_M 0 0 0 0
MULTI_F 0 0 0 0
MULTI_M 0 0 0 0
WHT_M 0 0 0 0

```

THETA

SPED AFAM_F AFAM_M AMIND_F AMIND_M

```

SPED 0
AFAM_F 0 0
AFAM_M 0 0 0
AMIND_F 0 0 0 0
AMIND_M 0 0 0 0 0
ASIAN_F 0 0 0 0 0
ASIAN_M 0 0 0 0 0
HAW_F 0 0 0 0 0
HAW_M 0 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0

```

THETA

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

```

ASIAN_F 0
ASIAN_M 0 0
HAW_F 0 0 0
HAW_M 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0

```

THETA

HISP_M MULTI_F MULTI_M WHT_M

```

HISP_M 0
MULTI_F 0 0
MULTI_M 0 0 0
WHT_M 0 0 0 0

```

ALPHA

SPED AFAM_F AFAM_M AMIND_F AMIND_M

0 0 0 0 0

ALPHA

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

0 0 0 0 0

ALPHA

HISP_M MULTI_F MULTI_M WHT_M

0 0 0 0

BETA

SPED AFAM_F AFAM_M AMIND_F AMIND_M

SPED 0 1 2 3 4
AFAM_F 0 0 0 0 0
AFAM_M 0 0 0 0 0
AMIND_F 0 0 0 0 0
AMIND_M 0 0 0 0 0
ASIAN_F 0 0 0 0 0
ASIAN_M 0 0 0 0 0
HAW_F 0 0 0 0 0
HAW_M 0 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0

BETA

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

SPED 5 6 7 8 9
AFAM_F 0 0 0 0 0
AFAM_M 0 0 0 0 0
AMIND_F 0 0 0 0 0
AMIND_M 0 0 0 0 0
ASIAN_F 0 0 0 0 0
ASIAN_M 0 0 0 0 0
HAW_F 0 0 0 0 0
HAW_M 0 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0

BETA

HISP_M MULTI_F MULTI_M WHT_M

SPED 10 11 12 13
AFAM_F 0 0 0 0
AFAM_M 0 0 0 0
AMIND_F 0 0 0 0
AMIND_M 0 0 0 0
ASIAN_F 0 0 0 0
ASIAN_M 0 0 0 0
HAW_F 0 0 0 0
HAW_M 0 0 0 0
HISP_F 0 0 0 0
HISP_M 0 0 0 0
MULTI_F 0 0 0 0
MULTI_M 0 0 0 0
WHT_M 0 0 0 0

PSI

SPED AFAM_F AFAM_M AMIND_F AMIND_M

SPED 0
AFAM_F 0 0
AFAM_M 0 0 0
AMIND_F 0 0 0 0
AMIND_M 0 0 0 0 0
ASIAN_F 0 0 0 0 0
ASIAN_M 0 0 0 0 0
HAW_F 0 0 0 0 0
HAW_M 0 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0

PSI

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

ASIAN_F 0
ASIAN_M 0 0
HAW_F 0 0 0
HAW_M 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0

PSI

HISP_M MULTI_F MULTI_M WHT_M

HISP_M 0
MULTI_F 0 0
MULTI_M 0 0 0
WHT_M 0 0 0 0

PARAMETER SPECIFICATION FOR BETWEEN

TAU

SPED\$1

15

NU

SPED

0

LAMBDA

SPED

SPED 0

THETA

SPED

SPED 0

ALPHA
SPED

0

BETA
SPED

SPED 0

PSI
SPED

SPED 14

STARTING VALUES FOR WITHIN

TAU
SPED\$1

0.000

NU
SPED AFAM_F AFAM_M AMIND_F AMIND_M

0.000 0.000 0.000 0.000 0.000

NU
ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

0.000 0.000 0.000 0.000 0.000

NU
HISP_M MULTI_F MULTI_M WHT_M

0.000 0.000 0.000 0.000

LAMBDA
SPED AFAM_F AFAM_M AMIND_F AMIND_M

SPED 1.000 0.000 0.000 0.000 0.000
AFAM_F 0.000 1.000 0.000 0.000 0.000
AFAM_M 0.000 0.000 1.000 0.000 0.000
AMIND_F 0.000 0.000 0.000 1.000 0.000
AMIND_M 0.000 0.000 0.000 0.000 1.000
ASIAN_F 0.000 0.000 0.000 0.000 0.000
ASIAN_M 0.000 0.000 0.000 0.000 0.000
HAW_F 0.000 0.000 0.000 0.000 0.000
HAW_M 0.000 0.000 0.000 0.000 0.000
HISP_F 0.000 0.000 0.000 0.000 0.000
HISP_M 0.000 0.000 0.000 0.000 0.000
MULTI_F 0.000 0.000 0.000 0.000 0.000
MULTI_M 0.000 0.000 0.000 0.000 0.000
WHT_M 0.000 0.000 0.000 0.000 0.000

LAMBDA

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

SPED 0.000 0.000 0.000 0.000 0.000
 AFAM_F 0.000 0.000 0.000 0.000 0.000
 AFAM_M 0.000 0.000 0.000 0.000 0.000
 AMIND_F 0.000 0.000 0.000 0.000 0.000
 AMIND_M 0.000 0.000 0.000 0.000 0.000
 ASIAN_F 1.000 0.000 0.000 0.000 0.000
 ASIAN_M 0.000 1.000 0.000 0.000 0.000
 HAW_F 0.000 0.000 1.000 0.000 0.000
 HAW_M 0.000 0.000 0.000 1.000 0.000
 HISP_F 0.000 0.000 0.000 0.000 1.000
 HISP_M 0.000 0.000 0.000 0.000 0.000
 MULTI_F 0.000 0.000 0.000 0.000 0.000
 MULTI_M 0.000 0.000 0.000 0.000 0.000
 WHT_M 0.000 0.000 0.000 0.000 0.000

LAMBDA

HISP_M MULTI_F MULTI_M WHT_M

SPED 0.000 0.000 0.000 0.000
 AFAM_F 0.000 0.000 0.000 0.000
 AFAM_M 0.000 0.000 0.000 0.000
 AMIND_F 0.000 0.000 0.000 0.000
 AMIND_M 0.000 0.000 0.000 0.000
 ASIAN_F 0.000 0.000 0.000 0.000
 ASIAN_M 0.000 0.000 0.000 0.000
 HAW_F 0.000 0.000 0.000 0.000
 HAW_M 0.000 0.000 0.000 0.000
 HISP_F 0.000 0.000 0.000 0.000
 HISP_M 1.000 0.000 0.000 0.000
 MULTI_F 0.000 1.000 0.000 0.000
 MULTI_M 0.000 0.000 1.000 0.000
 WHT_M 0.000 0.000 0.000 1.000

THETA

SPED AFAM_F AFAM_M AMIND_F AMIND_M

SPED 0.000
 AFAM_F 0.000 0.000
 AFAM_M 0.000 0.000 0.000
 AMIND_F 0.000 0.000 0.000 0.000
 AMIND_M 0.000 0.000 0.000 0.000 0.000
 ASIAN_F 0.000 0.000 0.000 0.000 0.000
 ASIAN_M 0.000 0.000 0.000 0.000 0.000
 HAW_F 0.000 0.000 0.000 0.000 0.000
 HAW_M 0.000 0.000 0.000 0.000 0.000
 HISP_F 0.000 0.000 0.000 0.000 0.000
 HISP_M 0.000 0.000 0.000 0.000 0.000
 MULTI_F 0.000 0.000 0.000 0.000 0.000
 MULTI_M 0.000 0.000 0.000 0.000 0.000
 WHT_M 0.000 0.000 0.000 0.000 0.000

THETA

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

ASIAN_F 0.000
 ASIAN_M 0.000 0.000
 HAW_F 0.000 0.000 0.000
 HAW_M 0.000 0.000 0.000 0.000
 HISP_F 0.000 0.000 0.000 0.000 0.000
 HISP_M 0.000 0.000 0.000 0.000 0.000

MULTI_F 0.000 0.000 0.000 0.000 0.000
MULTI_M 0.000 0.000 0.000 0.000 0.000
WHT_M 0.000 0.000 0.000 0.000 0.000

THETA

HISP_M MULTI_F MULTI_M WHT_M

HISP_M 0.000
MULTI_F 0.000 0.000
MULTI_M 0.000 0.000 0.000
WHT_M 0.000 0.000 0.000 0.000

ALPHA

SPED AFAM_F AFAM_M AMIND_F AMIND_M

0.000 0.000 0.000 0.000 0.000

ALPHA

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

0.000 0.000 0.000 0.000 0.000

ALPHA

HISP_M MULTI_F MULTI_M WHT_M

0.000 0.000 0.000 0.000

BETA

SPED AFAM_F AFAM_M AMIND_F AMIND_M

SPED 0.000 0.000 0.000 0.000 0.000
AFAM_F 0.000 0.000 0.000 0.000 0.000
AFAM_M 0.000 0.000 0.000 0.000 0.000
AMIND_F 0.000 0.000 0.000 0.000 0.000
AMIND_M 0.000 0.000 0.000 0.000 0.000
ASIAN_F 0.000 0.000 0.000 0.000 0.000
ASIAN_M 0.000 0.000 0.000 0.000 0.000
HAW_F 0.000 0.000 0.000 0.000 0.000
HAW_M 0.000 0.000 0.000 0.000 0.000
HISP_F 0.000 0.000 0.000 0.000 0.000
HISP_M 0.000 0.000 0.000 0.000 0.000
MULTI_F 0.000 0.000 0.000 0.000 0.000
MULTI_M 0.000 0.000 0.000 0.000 0.000
WHT_M 0.000 0.000 0.000 0.000 0.000

BETA

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

SPED 0.000 0.000 0.000 0.000 0.000
AFAM_F 0.000 0.000 0.000 0.000 0.000
AFAM_M 0.000 0.000 0.000 0.000 0.000
AMIND_F 0.000 0.000 0.000 0.000 0.000
AMIND_M 0.000 0.000 0.000 0.000 0.000
ASIAN_F 0.000 0.000 0.000 0.000 0.000
ASIAN_M 0.000 0.000 0.000 0.000 0.000
HAW_F 0.000 0.000 0.000 0.000 0.000
HAW_M 0.000 0.000 0.000 0.000 0.000
HISP_F 0.000 0.000 0.000 0.000 0.000
HISP_M 0.000 0.000 0.000 0.000 0.000
MULTI_F 0.000 0.000 0.000 0.000 0.000
MULTI_M 0.000 0.000 0.000 0.000 0.000

WHT_M 0.000 0.000 0.000 0.000 0.000

BETA

HISP_M MULTI_F MULTI_M WHT_M

SPED 0.000 0.000 0.000 0.000
AFAM_F 0.000 0.000 0.000 0.000
AFAM_M 0.000 0.000 0.000 0.000
AMIND_F 0.000 0.000 0.000 0.000
AMIND_M 0.000 0.000 0.000 0.000
ASIAN_F 0.000 0.000 0.000 0.000
ASIAN_M 0.000 0.000 0.000 0.000
HAW_F 0.000 0.000 0.000 0.000
HAW_M 0.000 0.000 0.000 0.000
HISP_F 0.000 0.000 0.000 0.000
HISP_M 0.000 0.000 0.000 0.000
MULTI_F 0.000 0.000 0.000 0.000
MULTI_M 0.000 0.000 0.000 0.000
WHT_M 0.000 0.000 0.000 0.000

PSI

SPED AFAM_F AFAM_M AMIND_F AMIND_M

SPED 1.000
AFAM_F 0.000 0.064
AFAM_M 0.000 0.000 0.066
AMIND_F 0.000 0.000 0.000 0.001
AMIND_M 0.000 0.000 0.000 0.000 0.001
ASIAN_F 0.000 0.000 0.000 0.000 0.000
ASIAN_M 0.000 0.000 0.000 0.000 0.000
HAW_F 0.000 0.000 0.000 0.000 0.000
HAW_M 0.000 0.000 0.000 0.000 0.000
HISP_F 0.000 0.000 0.000 0.000 0.000
HISP_M 0.000 0.000 0.000 0.000 0.000
MULTI_F 0.000 0.000 0.000 0.000 0.000
MULTI_M 0.000 0.000 0.000 0.000 0.000
WHT_M 0.000 0.000 0.000 0.000 0.000

PSI

ASIAN_F ASIAN_M HAW_F HAW_M HISP_F

ASIAN_F 0.010
ASIAN_M 0.000 0.010
HAW_F 0.000 0.000 0.000
HAW_M 0.000 0.000 0.000 0.000
HISP_F 0.000 0.000 0.000 0.000 0.040
HISP_M 0.000 0.000 0.000 0.000 0.000
MULTI_F 0.000 0.000 0.000 0.000 0.000
MULTI_M 0.000 0.000 0.000 0.000 0.000
WHT_M 0.000 0.000 0.000 0.000 0.000

PSI

HISP_M MULTI_F MULTI_M WHT_M

HISP_M 0.041
MULTI_F 0.000 0.010
MULTI_M 0.000 0.000 0.010
WHT_M 0.000 0.000 0.000 0.087

STARTING VALUES FOR BETWEEN

TAU
 SPED\$1

 0.915

NU
 SPED

 0.000

LAMBDA
 SPED

 SPED 1.000

THETA
 SPED

 SPED 0.000

ALPHA
 SPED

 0.000

BETA
 SPED

 SPED 0.000

PSI
 SPED

 SPED 1.000

PRIORS FOR ALL PARAMETERS PRIOR MEAN PRIOR VARIANCE PRIOR STD. DEV.

Parameter 1~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 2~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 3~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 4~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 5~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 6~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 7~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 8~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 9~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 10~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 11~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 12~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 13~N(0.000,5.000) 0.0000 5.0000 2.2361
 Parameter 14~IG(-1.000,0.000) infinity infinity infinity
 Parameter 15~N(0.000,5.000) 0.0000 5.0000 2.2361

TECHNICAL 8 OUTPUT

TECHNICAL 8 OUTPUT FOR BAYES ESTIMATION

PROCESSOR BSEED

1 0
2 285380
3 253358
4 93468

POTENTIAL PARAMETER WITH

ITERATION SCALE REDUCTION HIGHEST PSR

100 1.325 5
200 1.041 7
300 1.048 14
400 1.213 5
500 1.104 5
600 1.023 5
700 1.033 14
800 1.016 6
900 1.015 3
1000 1.025 3
1100 1.021 3
1200 1.017 3
1300 1.021 5
1400 1.031 5
1500 1.019 5
1600 1.011 5
1700 1.024 5
1800 1.032 6
1900 1.035 5
2000 1.033 6
2100 1.027 6
2200 1.029 6
2300 1.028 6
2400 1.022 14
2500 1.018 14
2600 1.018 14
2700 1.017 14
2800 1.013 14
2900 1.008 14
3000 1.008 5
3100 1.009 5
3200 1.010 5
3300 1.009 5
3400 1.009 5
3500 1.009 5
3600 1.012 5
3700 1.015 5
3800 1.010 5
3900 1.008 5
4000 1.010 5
4100 1.007 5
4200 1.004 5
4300 1.002 5
4400 1.002 5
4500 1.001 4
4600 1.002 6
4700 1.003 6
4800 1.003 4
4900 1.003 4
5000 1.002 6

DIAGRAM INFORMATION

Mplus diagrams are currently not available for multilevel analysis.
No diagram output was produced.

Beginning Time: 20:03:42

Ending Time: 20:35:46
Elapsed Time: 00:32:04

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